

CSD-334: USER INTERFACE DESIGN

UX Engineering: Performance & Testing

Deep Dive into Offline-First Architecture and A/B Testing Methodologies

Assignment #1 | Chapters 3.3 & 3.4



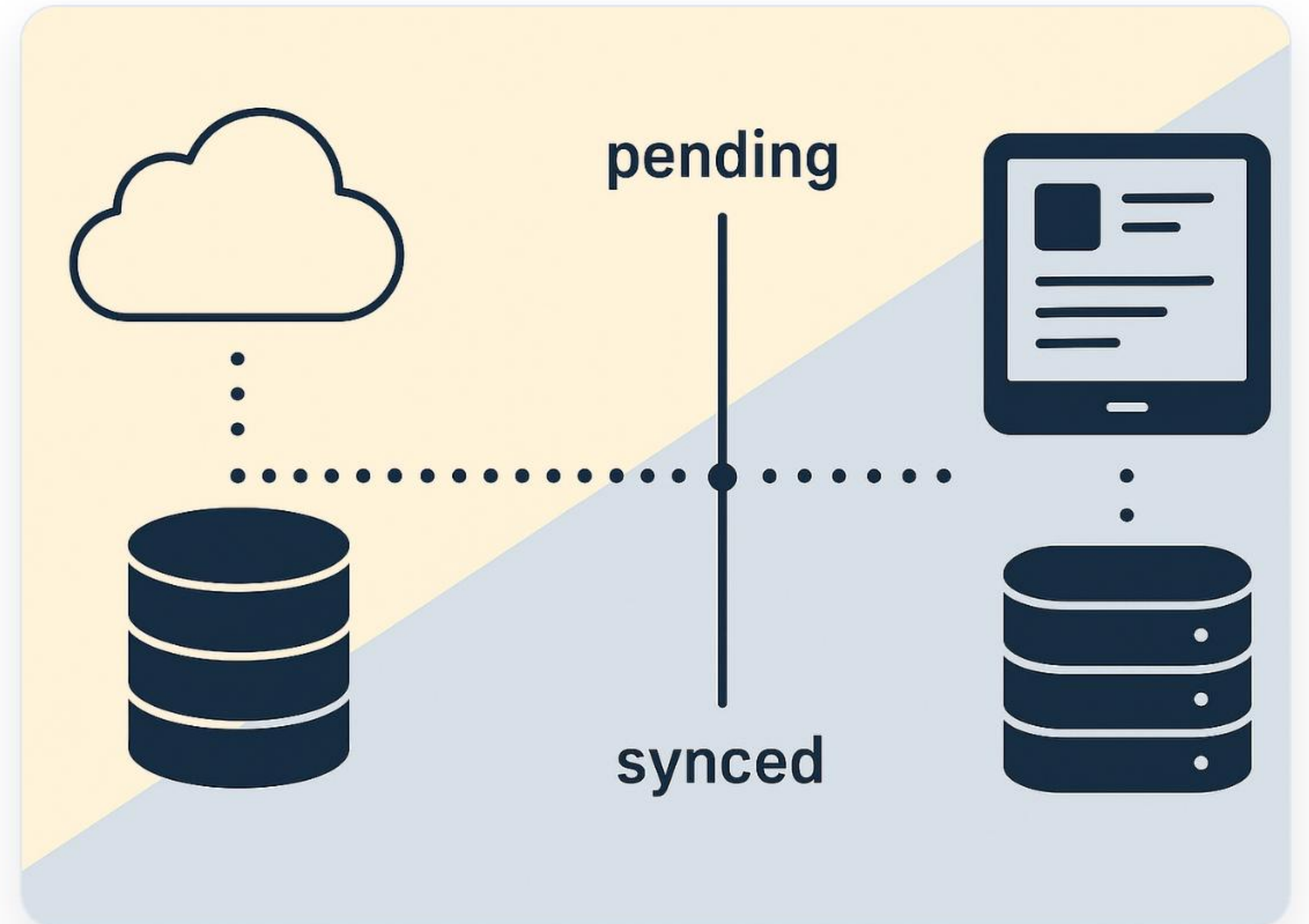
Performance Considerations

Section 3.3 Focus: **Offline-First Applications**

| Defining Offline-First Design

Traditional apps assume **constant connectivity**. If the internet fails, the app fails. **Offline-First** flips this paradigm.

- **The Paradigm:** Treats a lack of network not as an error, but as a standard state.
- **Source of Truth:** The local database is the primary source of truth, not the server.
- **User Perception:** Users can interact with the app anytime, anywhere, with zero loading states.



| How It Works: Sync Architecture



Local Storage

Uses Room, SQLite, or IndexedDB to store all data locally on the device.



Sync Engine

Automatically detects network and reconciles local changes with the server.



Conflict Logic

Handles cases where data changed on both the



Perceived Speed

UI updates instantly
(Optimistic Updates)

Key Logic

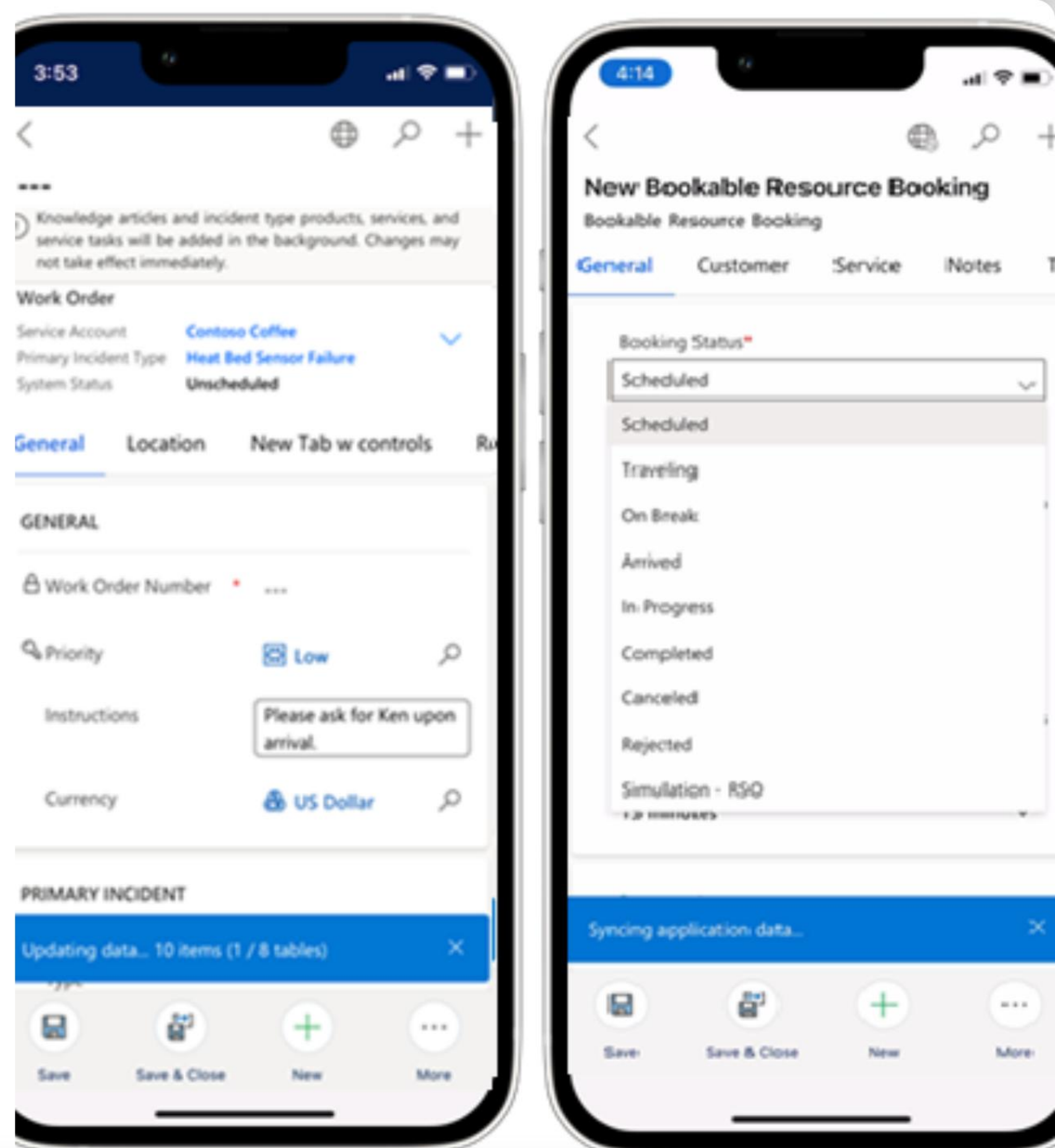
```
if (Network.isAvailable()) {  
    SyncQueue.push();  
    UpdateServer();  
} else {  
    SaveToLocalStorage();  
    MarkAsPending();  
}
```

This ensures a "seamless transition" between online and offline states.

Example: Instagram Feed

Instagram uses a sophisticated **Offline-First approach** to ensure user engagement remains high even in low-signal areas.

- ✓ **Pre-caching:** Loads images while you are scrolling, storing them for offline viewing later.
- ✓ **Background Posting:** When you post a photo offline, it "appears" to be posted, then syncs silently once signal returns.

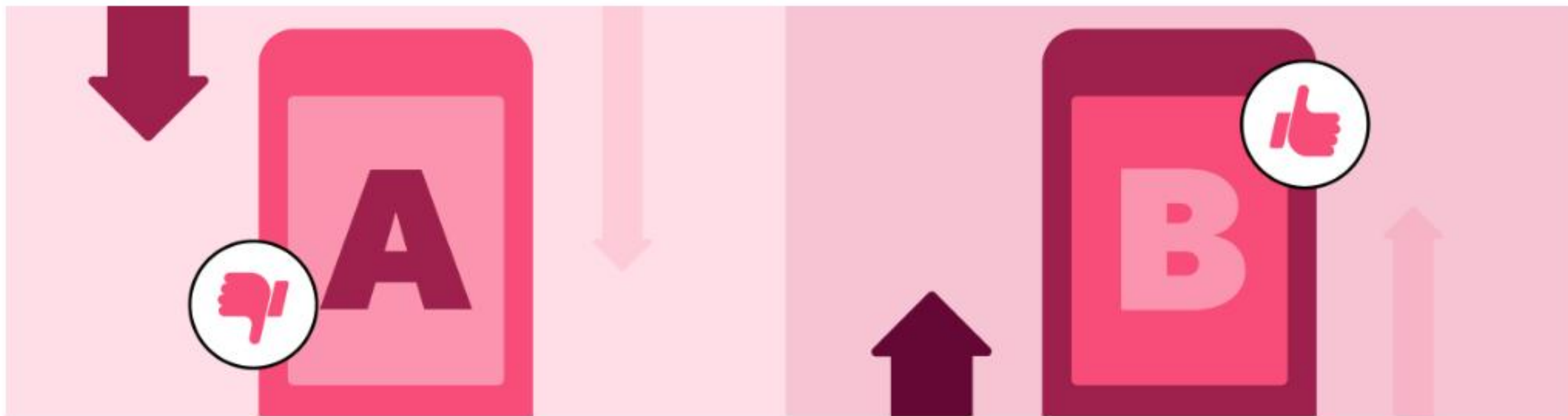




UX Testing & Prototyping

Section 3.4 Focus: **A/B Testing Methodology**

| What is A/B Testing?

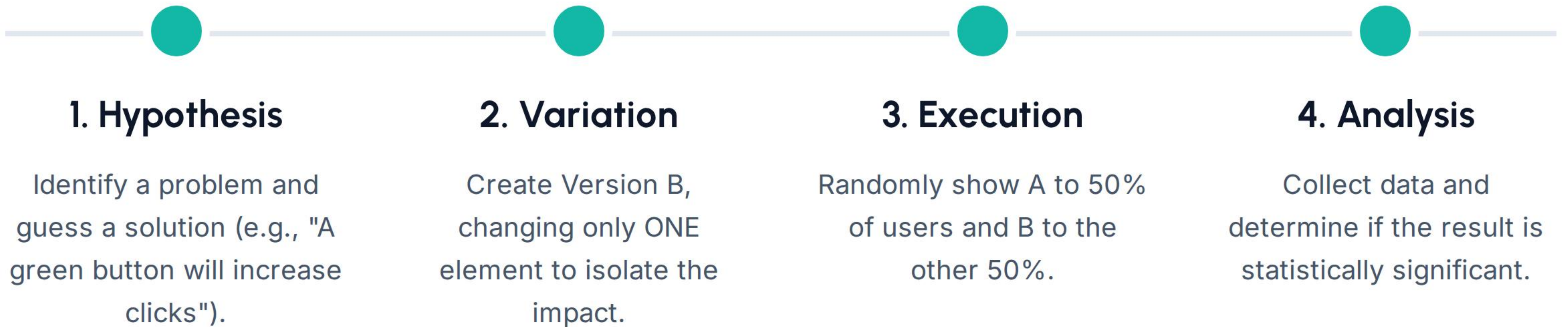


A/B testing (or split testing) is a **quantitative research method** where two versions of a design are compared to see which performs better.

- **Version A (Control):** The current design/original version.
- **Version B (Variant):** The new design with one specific change.
- **Goal:** Replace opinions with **real user data**.

"If you don't measure it, you can't improve it." — Peter Drucker

| The A/B Testing Workflow



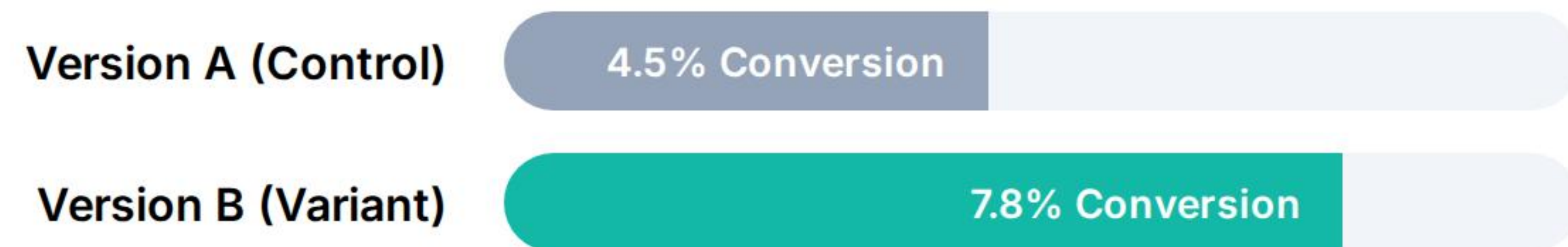
| Key UX Metrics to Measure

Metric	Definition	Ideal for Testing...
Conversion Rate	% of users who complete a desired action (e.g., purchase).	Checkouts, Sign-up forms
Click-Through Rate (CTR)	% of users who click a specific link or button.	CTA Buttons, Navigation menus
Bounce Rate	% of users who leave the page without interacting.	Landing page layouts, Headings
Time on Task	Average time spent to complete a specific flow.	User efficiency, Complexity

| Visualizing Results: A Case Study

Results Summary

In this test, we changed the "Buy Now" button from **Gray** to **Teal**.



Verdict: Version B resulted in a 73% increase in sales.

📈 Statistical Significance

Before implementing B, we must ensure the p-value is < 0.05 . This confirms the results weren't just luck or random chance.

- Confidence Level: 95%
- Sample Size: 10,000 Users
- Duration: 14 Days

| Key Takeaways

Performance

Offline-first design is essential for modern mobility, ensuring apps remain functional regardless of signal strength.

UX Testing

A/B Testing provides an objective, data-driven framework for making UI decisions that actually work.

Iteration

Great UX is never "finished." It is an iterative cycle of designing, measuring, and refining based on real behavior.

References & Sources

📖 **Chapter 3.3 & 3.4:** Course Material on Performance & Testing.

🌐 **Nielsen Norman Group (NN/G):** "A/B Testing 101" - Best practices for UX research.

💻 **Google PWA Docs:** "Offline-First Architectures" and Service Worker patterns.

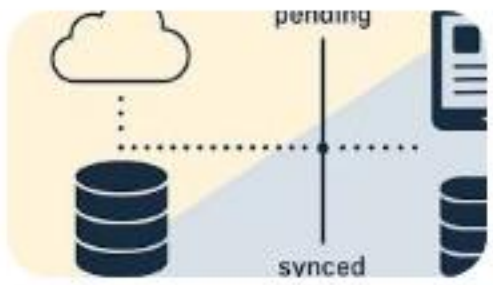
📱 **Instagram Engineering Blog:** "Making Instagram work offline".



Thank You!

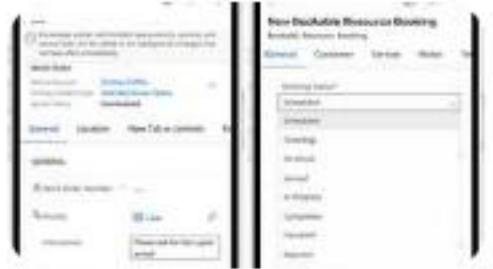
Any Questions?

| Image Sources



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